

Question

The element **M** of the first transition series dissolves in dilute sulfuric acid and crystallizes to obtain a hydrated sulfate crystal **A**, in which the mass fraction of **M** is 20.97%. Element **M** has only one non-radioactive stable isotope.

The THF solvate of the simple chloride of **M** reacts with 1-lithiated bicyclo[2.2.1]heptane in pentane to produce **M** element and a mononuclear, electrically neutral complex **D** with a carbon content of 76.50%.

Reduction of **D** with lithium element in tetrahydrofuran yields a salt **E** with a cation-anion ratio of 1:1, while oxidation of **D** with silver tetrafluoroborate yields a salt **F** with a cation-anion ratio of 1:1.

Regarding the above content, there are the following conclusions:

Conclusion 1: Isotope M_2 whose mass number is 1 greater than that of the stable isotope of **M** emits β rays when decaying.

Conclusion 2: The molecular weight of **D** is 583.77.

Conclusion 3: **D**, **E**, and **F** all contain **M** with an octahedral coordination configuration.

Conclusion 4: The theoretically calculated spin-only magnetic moment of **E** is 2.83 B.M..

Conclusion 5: **F** is a diamagnetic substance.

Select the option corresponding to all the correct conclusions.

A. All other options are incorrect

B. Option 1, Option 2, Option 3

C. Option 1, Option 2, Option 4

D. Option 1, Option 2, Option 5

E. Option 1, Option 3, Option 4

F. Option 1, Option 3, Option 5

G. Option 1, Option 4, Option 5

H. Option 2, Option 3, Option 4

I. Option 2, Option 3, Option 5

J. Option 2, Option 4, Option 5

K. Option 3, Option 4, Option 5

Answer

Correct Answer: **G**

Detailed Explanation

Only *Sc*, *Mn*, and *Co* are elements in the first transition series that have only one stable isotope. Examine the mass fraction of the metal in their hydrated sulfates as follows:

- If **M** is *Sc*: *Sc* usually forms +3 valent ions in acid, so the salt is $Sc_2(SO_4)_3 \cdot nH_2O$. The relative molecular mass $M_r(\mathbf{A}) = 2 \times 44.96 \div 0.2097 = 428.8$, the number of moles of water $n = (428.8 - (2 \times 44.96 + 3 \times (32.07 + 4 \times 16.00))) \div 18.02 = 2.81$, which is not an integer, so *Sc* is excluded.

- If **M** is *Mn*: *Mn* usually forms +2 valent ions in dilute sulfuric acid, so the salt is $MnSO_4 \cdot nH_2O$. $M_r(\mathbf{A}) = 54.94 \div 0.2097 = 262.0$, the number of moles of water $n = (262.0 - (54.94 + 32.07 + 4 \times 16.00)) \div 18.02 = 6.16$, which is not an integer, so *Mn* is excluded.

- If **M** is *Co*: *Co* forms +2 valent ions in dilute sulfuric acid, so the salt is $CoSO_4 \cdot nH_2O$. $M_r(\mathbf{A}) = 58.93 \div 0.2097 = 281.0$, the number of moles of water $n = (281.0 - (58.93 + 32.07 + 4 \times 16.00)) \div 18.02 = 7.00$, which is an integer, so it may be *Co*.

Therefore, the metal **M** is cobalt (*Co*), and its only stable isotope is ^{59}Co .

CHECKPOINT

3 PTS

The metal element **M** is cobalt

The only simple chloride of *Co* commonly used for synthesis is $CoCl_2$. The reaction of $CoCl_2$ with 1-lithiated bicyclo[2.2.1]heptane (denoted as Li-R) to produce *Co* element indicates that *Co* may have undergone a

disproportionation reaction, simultaneously producing Co element and CoR_x . **D** may be CoR_x , with the molecular formula $CoC_{7x}H_{11x}$. List the mass fraction equation:

$$\frac{7x \times 12.01}{58.93 + 7x \times 12.01 + 11x \times 1.008} = 76.50\%$$

Solving for: $x = 3.999 \approx 4$, so **D** is CoR_4 .

CHECKPOINT

3 PTS

D is CoR_4

Lithium element is a strong reducing agent, and the product **E** formed by reducing **D** should be a lithium salt with **D** carrying several negative charges as the anion. According to the 1:1 ratio of cations to anions, it can be inferred that **E** is $[Li(THF)_x][CoR_4]$

CHECKPOINT

1 PTS

E is $[Li(THF)_x][CoR_4]$

+1 valent silver ion is a moderately strong oxidizing agent, and the product **F** formed by $AgBF_4$ oxidizing **D** should be a tetrafluoroborate salt with **D** carrying several positive charges as the cation. According to the 1:1 ratio of cations to anions, it can be inferred that **F** is $[CoR_4][BF_4]$

CHECKPOINT

1 PTS

F is $[CoR_4][BF_4]$

The isotope **M**₂ whose mass number is 1 greater than the stable isotope of **M** is ^{60}Co . ^{60}Co undergoes β decay and emits β rays. Conclusion 1 is correct.

CHECKPOINT

1 PTS

The isotope **M**₂ whose mass number is 1 greater than the stable isotope of **M** is ^{60}Co . ^{60}Co undergoes β decay and emits β rays.

The molecular weight of **D**: $M_r(\mathbf{D}) = 58.93 + 4 \times (7 \times 12.01 + 11 \times 1.008) = 439.56 \neq 583.77$. Conclusion 2 is incorrect.

CHECKPOINT

1 PTS

The molecular weight of **D**: $M_r(\mathbf{D}) = 58.93 + 4 \times (7 \times 12.01 + 11 \times 1.008) = 439.56 \neq 583.77$

Co in **D**, **E**, and **F** all have a tetrahedral coordination configuration. Conclusion 3 is incorrect.

CHECKPOINT

1 PTS

Co in **D**, **E**, and **F** all have a tetrahedral coordination configuration.

Co in **E** is +3 valent, and its electronic configuration in a tetrahedral field is $(e)^4(t_2)^2$, with 2 unpaired electrons. The theoretically calculated spin-only magnetic moment of **E** is $\sqrt{2 \times (2 + 2)} = 2.83$ B. M. Conclusion 4 is correct.

CHECKPOINT

1 PTS

Co in **E** is +3 valent, and its electronic configuration in a tetrahedral field is $(e)^4(t_2)^2$, with 2 unpaired electrons. The theoretically calculated spin-only magnetic moment of **E** is $\sqrt{2 \times (2 + 2)} = 2.83$ B. M.

Co in **F** is +5 valent, and its electronic configuration in a tetrahedral field is $(e)^4(t_2)^0$, with no unpaired electrons, so it is diamagnetic. Conclusion 5 is correct.

CHECKPOINT

1 PTS

Co in **F** is +5 valent, and its electronic configuration in a tetrahedral field is $(e)^4(t_2)^0$, with no unpaired electrons, so it is diamagnetic.

In summary, choose option G.